Question 1

While researching a topic, a student has taken the following notes:

- Organisms release cellular material into their environment by shedding substances such as hair or skin.
- The DNA in these substances is known as environmental DNA, or eDNA.
- Researchers collect and analyze eDNA to detect the presence of species that are difficult to observe.
- Geneticist Sara Oyler-McCance's research team analyzed eDNA in water samples from the Florida Everglades to detect invasive constrictor snake species in the area.
- The study determined a 91% probability of detecting Burmese python eDNA in a given location.

The student wants to present the study to an audience already familiar with environmental DNA. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Sara Oyler-McCance's researchers analyzed eDNA in water samples from the Florida Everglades for evidence of invasive constrictor snakes, which are difficult to observe.

B. An analysis of eDNA can detect the presence of invasive species that are difficult to observe, such as constrictor snakes.

C. Researchers found Burmese python eDNA, or environmental DNA, in water samples; eDNA is the DNA in released cellular materials, such as shed skin cells.

D. Sara Oyler-McCance's researchers analyzed environmental DNA (eDNA)—that is, DNA from cellular materials released by organisms—in water samples from the Florida Everglades.

Question 2

While researching a topic, a student has taken the following notes:

- Allan Houser was a Chiricahua Warm Springs Apache sculptor, illustrator, and painter.
- Many of his sculptures featured Native American figures.
- He depicted this subject matter using abstract, modernist forms, developing a distinctive style that influenced many other artists.
- His well-known sculpture *Sacred Rain Arrow* was pictured on the State of Oklahoma license plate.

The student wants to describe the distinctive style of Houser's sculptures. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. A sculptor, illustrator, and painter, Houser developed a distinctive style for portraying Native American figures.

B. Houser's sculptures employ abstract, modernist forms to depict Native American figures.

C. Many other artists have been influenced by the style of Houser's sculptures.

D. The sculpture *Sacred Rain Arrow* is a well-known example of Houser's style.

Question 3

While researching a topic, a student has taken the following notes:

- John Carver was one of the 41 signatories of the Mayflower Compact.
- The Mayflower Compact was a legal agreement among the pilgrims that immigrated to Plymouth Colony.
- It was created in 1620 to establish a common government.
- It states that the pilgrims who signed it wanted to "plant the first colony in the northern parts of Virginia" under King James.
- Carver became the first governor of Plymouth Colony.

The student wants to specify the reason the Mayflower Compact was created. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Stating that its signatories wanted to "plant the first colony in the northern parts of Virginia," the Mayflower Compact was a legal agreement among the pilgrims that immigrated to Plymouth Colony.

B. Created in 1620, the Mayflower Compact states that the pilgrims wanted to "plant the first colony in the northern parts of Virginia."

C. The Mayflower Compact was created to establish a common government among the pilgrims that immigrated to Plymouth Colony.

D. The Mayflower Compact had 41 signatories, including John Carver, the first governor of Plymouth Colony.

Question 4

While researching a topic, a student has taken the following notes:

- Mexican tetras are a fish species with two distinct populations.
- Surface-dwelling tetras live on the surface and are able to see.
- Cave-dwelling tetras live in total darkness and have lost the ability to see.
- Cave-dwelling tetras have asymmetrical skulls with more sensory receptors on one side than the other.
- These receptors help cave-dwelling tetras navigate in darkness.

The student wants to emphasize a difference between surface-dwelling and cave-dwelling tetras. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Surface-dwelling and cave-dwelling tetras may belong to the same species, but they are quite different.

B. Cave-dwelling tetras can no longer see but use sensory receptors on their skulls to navigate.

C. Mexican tetras are a fish species with two distinct populations: surface-dwelling tetras and cave-dwelling tetras.

D. Surface-dwelling tetras can see, whereas cave-dwelling tetras cannot.

Question 5

While researching a topic, a student has taken the following notes:

- A marathon is a long-distance running race that is 26.2 miles long.
- An ultramarathon is a long-distance running race of more than 26.2 miles.
- The Kepler Challenge is a one-day, 37.3-mile ultramarathon in New Zealand.
- The Spreelauf is a six-day, 261-mile ultramarathon in Germany.

The student wants to make a generalization about ultramarathons. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Examples of ultramarathons include the 37.3-mile Kepler Challenge in New Zealand and the 261-mile Spreelauf in Germany.

B. A marathon is 26.2 miles long, but the Spreelauf ultramarathon, at 261 miles, is far longer.

C. Ultramarathons range widely in length, from a few dozen miles to a few hundred.

D. While the Kepler Challenge is a one-day ultramarathon, the Spreelauf is a six-day ultramarathon.

Question 6

While researching a topic, a student has taken the following notes:

- From Earth, all the meteors in a meteor shower appear to originate from a single spot in the sky.
- This spot is called the meteor shower's radiant.
- The Perseid meteor shower is visible in the northern hemisphere in July and August.
- Like many meteor showers, it is named for the location of its radiant.
- Its radiant is located within the constellation Perseus.

The student wants to explain the origin of the Perseid meteor shower's name. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. The Perseid meteor shower is named for the constellation Perseus, the location of the meteor shower's radiant.

B. A meteor shower's name may be linked to a single spot in the sky.

C. The Perseid meteor shower, which has a radiant, is visible in the northern hemisphere in July and August.

D. From Earth, all the meteors in a meteor shower appear to originate from a radiant, such as the one within Perseus.

Question 7

While researching a topic, a student has taken the following notes:

- Architect Julian Abele studied Gregorian and neo-Gothic architecture in Europe.
- Abele worked for an architecture firm that was hired in 1924 to design buildings for Duke University's new campus.
- Most of the buildings on Duke's campus were designed in the Gregorian or neo-Gothic architectural styles.
- At the time, Abele was not formally credited with designing the buildings.
- Based on the buildings' architectural styles, historians believe Abele designed most of the campus buildings.

The student wants to specify why historians believe Abele designed most of Duke's campus buildings. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Given that most of the buildings on Duke's campus feature architectural styles that Abele had studied in Europe, historians believe Abele is the one who designed them.

B. Though Abele wasn't formally credited at the time, historians believe he designed most of the buildings on Duke's campus.

C. Most of Duke's campus buildings, which were designed by a firm Abele worked for, were designed in the Gregorian and neo-Gothic architectural styles.

D. Abele, an architect who studied Gregorian and neo-Gothic architecture in Europe, is believed to have designed most of the buildings on Duke's campus.

Question 8

While researching a topic, a student has taken the following notes:

- J.R.R. Tolkien's 1937 novel *The Hobbit* features two maps.
- The novel opens with a reproduction of the map that the characters use on their quest.
- This map introduces readers to the fictional world they are about to enter.
- The novel closes with a map depicting every stop on the characters' journey.
- That map allows readers to reconstruct the story they have just read.

The student wants to contrast the purposes of the two maps in *The Hobbit*. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. *The Hobbit*'s opening map introduces readers to the fictional world they are about to enter, while the closing map allows them to reconstruct the story they have just read.

B. *The Hobbit*, a novel published by J.R.R. Tolkien in 1937, features a reproduction of a map that the characters use on their quest, as well as a map that appears at the end of the novel.

C. *The Hobbit*'s two maps, one opening and one closing the novel, each serve a purpose for readers.

D. In 1937, author J.R.R. Tolkien published *The Hobbit*, a novel featuring both an opening and a closing map.

Question 9

While researching a topic, a student has taken the following notes:

- In the art world, the term biennial traditionally refers to an art exhibition that takes place every two years in a single location.
- Such biennials are held in New York, Berlin, and Venice.
- In 2006, artists Ed Gomez and Luis Hernandez founded the unconventional MexiCali Biennial.
- The MexiCali Biennial hosts exhibitions in different venues on both sides of the US-Mexico border.
- The MexiCali Biennial has taken place on an uneven schedule, with exhibitions in 2006, 2009–10, 2013, and 2018–20.

The student wants to emphasize a difference between the MexiCali Biennial and traditional biennials. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. In 2006, artists Ed Gomez and Luis Hernandez founded the MexiCali Biennial, which has taken place in 2006, 2009–10, 2013, and 2018–20.

B. Unlike traditional biennials, the MexiCali Biennial hosts exhibitions in different venues on an uneven schedule.

C. The term biennial traditionally refers to an art exhibition that takes place every two years in a single location, not to exhibitions hosted at a variety of times and venues.

D. Biennial exhibitions have been held in New York, Berlin, and Venice but also on both sides of the US-Mexico border.

Question 10

While researching a topic, a student has taken the following notes:

- The Pueblo of Zuni is located about 150 miles west of Albuquerque, New Mexico.
- It is the traditional home of the A:shiwi (Zuni) people.
- The A:shiwi A:wan Museum and Heritage Center was established by tribal members in 1992.
- Its mission is stated on its website: "As a tribal museum and heritage center for the Zuni people and by the Zuni people we work to provide learning experiences that emphasize A:shiwi ways of knowing, as well as exploring modern concepts of knowledge and the transfer of knowledge."

The student wants to emphasize how long the museum has existed. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. The Pueblo of Zuni is home to the A:shiwi A:wan Museum and Heritage Center, which was founded by tribal members.

B. The A:shiwi A:wan Museum and Heritage Center has served the Pueblo of Zuni since 1992.

C. According to its website, the A:shiwi A:wan Museum and Heritage Center (founded in the 1990s) works to "emphasize A:shiwi ways of knowing."

D. Knowledge has been one of the central themes of the A:shiwi A:wan Museum and Heritage Center from its founding.

Question 11

While researching a topic, a student has taken the following notes:

- Just like states have state flags, some cities have city flags.
- Over one hundred US cities have redesigned their flags since 2015.
- The city of Pocatello, Idaho, redesigned its flag after it was named the most poorly designed flag in North America.
- Pocatello's new flag better represents the city's mountainous geography and civic priorities.
- Residents consider the new flag to be a meaningful symbol of civic pride.

The student wants to make and support a generalization about the effect of redesigning a city flag. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Over one hundred US cities have redesigned their flags, including Pocatello, whose flag had been named the most poorly designed flag in North America.

B. Pocatello is just one of over one hundred US cities that have redesigned their flags.

C. After it was named the most poorly designed flag in North America, the flag of Pocatello was redesigned to better represent the city's geography and civic priorities.

D. Redesigning a poorly designed city flag can create a meaningful symbol of civic pride, as was the case when Pocatello redesigned its original flag to better represent its geography and civic priorities.

Question 12

While researching a topic, a student has taken the following notes:

- Minnesota defines a lake as an inland body of water of at least 10 acres.
- Wisconsin's definition of a lake doesn't take size into account.
- By its own definition, Wisconsin has over 15,000 lakes, many smaller than 10 acres.
- By Minnesota's definition, Wisconsin has only about 6,000 lakes.

The student wants to contrast Minnesota's definition of a lake with Wisconsin's. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Wisconsin, which doesn't take size into account in defining a lake, claims that it has over 15,000 lakes.

B. Because its definition of a lake is different from Minnesota's, it is unclear how many lakes Wisconsin really has.

C. According to Minnesota's definition of a lake—an inland body of water of at least 10 acres—Wisconsin has about 6,000 lakes.

D. Minnesota's definition of a lake—an inland body of water of at least 10 acres—is more restrictive than Wisconsin's, which doesn't take size into account.

Question 13

While researching a topic, a student has taken the following notes:

In meteorology, an air mass is a large body of air with generally uniform humidity and temperature.

Air masses are commonly classified by two-letter names.

The first letter indicates the humidity of the air mass, while the second letter indicates the temperature.

cA (continental arctic) means dry and cold, for example. mT (maritime tropical) means moist and warm.

This classification system is based on the work of a Swedish meteorologist named Tor Bergeron (1891–1977).

The student wants to provide an example of an air mass. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Air masses are large bodies of air with generally uniform humidity and temperature.

B. The air mass classification system uses two-letter names and is based on the work of Tor Bergeron, a Swedish meteorologist.

C. Air masses are commonly classified by a two-letter name that indicates humidity and temperature.

D. One type of air mass is known as a cA, or continental arctic, air mass because it is dry and cold.

Question 14

While researching a topic, a student has taken the following notes:

The international Slow Food movement was founded in 1989 with the signing of the "Slow Food Manifesto."

The movement promotes universal access to healthy, high-quality food.

It calls for sustainable food production practices that protect local environments, ecosystems, and biodiversity.

It advocates for fair treatment of and compensation for food production workers.

The Slow Food USA organization was founded in 2000.

The student wants to introduce the Slow Food movement to a new audience. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. The international Slow Food movement, founded in 1989, promotes universal access to healthy, high-quality food that is produced sustainably by workers who are treated and compensated fairly.

B. The signing of the "Slow Food Manifesto" in 1989 marked the founding of the international Slow Food movement, while the Slow Food USA organization was founded in 2000.

C. The Slow Food movement advocates for food production workers.

D. Goals of the movement include universal access to healthy, high-quality food and sustainable food practices.

Question 15

While researching a topic, a student has taken the following notes:

In the early 1900s, suffragists organized marches for women's voting rights.
Suffragists in the United Kingdom marched from Edinburgh to London.
This march began on October 12, 1912, and ended on November 16, 1912.
Suffragists in the United States marched from New York City to Albany, New York.
This march began on December 16, 1912, and ended on December 28, 1912.

The student wants to emphasize the order in which the two marches occurred. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. After suffragists in the UK marched from Edinburgh to London, suffragists in the US marched from New York City to Albany, New York.

B. In the early 1900s, suffragists in the UK and the US marched for women's voting rights.

C. A march from New York City to Albany, New York, was followed by one that began in Edinburgh and ended in London.

D. From October 12 to November 16, 1912, suffragists in the UK marched from Edinburgh to London.

Question 16

While researching a topic, a student has taken the following notes:

Mary Kang is a Korean American portrait photographer. She is based in New York City and in Austin, Texas.
One of Kang's photographs features artist Dominique Fung. In the portrait, Fung is seated on the floor.
Five of Fung's paintings are resting against the wall behind her.

The student wants to describe where Fung is in the photograph to an audience already familiar with Kang and Fung. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Dominique Fung is in a photograph by Mary Kang, a portrait photographer based in New York City and Austin, Texas.

B. Mary Kang is a photographer based in both New York City and Austin, Texas.

C. Five paintings by artist Dominique Fung can be seen in the background of Mary Kang's photograph.

D. In Kang's portrait of her, Fung is seated on the floor, with five of her paintings resting against the wall behind her.

Question 17

While researching a topic, a student has taken the following notes:

Pointillism is a painting technique in which small, distinct dots of color are applied in patterns to form an image.
Betty Acquah is an artist from Ghana who uses pointillism in her work.
"By extending dabs of color in the subject matter into the background and vice-versa, an illusion of movement is created," she says about pointillism.
Her work often portrays Ghanaian women, whom she sees as the "unsung heroines of the Ghanaian Republic."
Her pointillist painting "Exquisite" (2016) features five dancing women twirling their skirts.

The student wants to provide a quotation from Acquah that explains why she used pointillism in "Exquisite." Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. In painting "Exquisite," Acquah applied pointillism to create what she called an "illusion of movement" within the painting's five dancing women and their twirling skirts.

B. Pointillism, the technique used in Acquah's "Exquisite," involves the application of small, distinct dots of color.

C. In "Exquisite," Acquah uses a technique that she says involves "extending dabs of color in the subject matter into the background and vice-versa."

D. "Exquisite" portrays Acquah's fellow Ghanaian women as she sees them: the "unsung heroes of the Ghanaian Republic."

Question 18

While researching a topic, a student has taken the following notes:

Thailand's annual Songkran Water Festival is held each April. It marks Songkran, the traditional Thai New Year.

People splash and spray each other for fun at the festival's community-wide water fights.

In Bangkok, thousands gather along Silom Road for the city's largest water fight.

In Chiang Mai, thousands gather at a historical monument called the Tha Phae Gate for the city's largest water fight.

The student wants to emphasize a similarity in how people in Bangkok and Chiang Mai celebrate Songkran. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. The largest water fight in Bangkok takes place along a city street, whereas the largest water fight in Chiang Mai takes place at a historical monument.

B. In both Bangkok and Chiang Mai, thousands gather to celebrate Songkran with water fights.

C. People in both Bangkok and Chiang Mai celebrate Songkran, but they don't do so in exactly the same way.

D. Each April, people in Thailand celebrate Songkran, the traditional Thai New Year.

Question 19

While researching a topic, a student has taken the following notes:

In 2022, University of Miami researchers discovered brine pools in the Gulf of Aqaba.

A brine pool is an underwater lake that sits on the ocean floor.

The water in brine pools is three to eight times saltier than the surrounding ocean.

The extreme saltiness of this water makes it toxic to most sea life.

Some forms of bacteria are able to survive in brine pools.

The student wants to explain why brine pools are toxic to most sea life. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Though brine pools are toxic to most sea life, some bacteria can survive there.

B. The water in brine pools is toxic to most sea life because it is three to eight times saltier than the surrounding ocean.

C. The brine pools in the Gulf of Aqaba are toxic to most sea life and were discovered by researchers in 2022.

D. Brine pools are salty underwater lakes that sit on the ocean floor.

Question 20

While researching a topic, a student has taken the following notes:

Crown shyness is a phenomenon in which the tops (crowns) of neighboring trees grow close together but don't overlap.

To explain how this happens, Australian forester M.R. Jacobs proposes the mutual abrasion theory.

According to Jacobs's theory, when trees brush against one another, branches break off.

Malaysian scholar Francis S.P. Ng posits the mutual shade avoidance theory.

According to Ng's theory, when tree branches detect shade from nearby trees' branches, they stop growing.

The student wants to compare the causes of crown shyness proposed in the two theories. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. While Jacobs proposes that crown shyness is caused by neighboring tree branches brushing against one another, Ng posits that it occurs when branches detect shade from nearby trees' branches.

B. Both Jacobs and Ng have proposed theories to explain what causes crown shyness.

C. Ng posits the mutual shade avoidance theory, whereas Jacobs proposes an alternative theory.

D. Jacobs's mutual abrasion theory proposes that when neighboring trees brush against one another, branches break off, resulting in a phenomenon in which the tops of trees grow close together but don't overlap.

Question 21

While researching a topic, a student has taken the following notes:

- In the midst of the US Civil War, Susie Taylor escaped slavery and fled to Union-army-occupied St. Simons Island off the Georgia coast.
- She began working for an all-Black army regiment as a nurse and teacher.
- In 1902, she published a book about the time she spent with the regiment.
- Her book was the only Civil War memoir to be published by a Black woman.
- It is still available to readers in print and online.

The student wants to emphasize the uniqueness of Taylor's accomplishment. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Taylor fled to St. Simons Island, which was then occupied by the Union army, for whom she began working.

B. After escaping slavery, Taylor began working for an all-Black army regiment as a nurse and teacher.

C. The book Taylor wrote about the time she spent with the regiment is still available to readers in print and online.

D. Taylor was the only Black woman to publish a Civil War memoir.

Question 22

While researching a topic, a student has taken the following notes:

- Sam Maloof (1916–2009) was an American woodworker and furniture designer.
- He was the son of Lebanese immigrants.
- He received a "genius grant" from the John D. and Catherine T. MacArthur Foundation in 1985.
- The Museum of Fine Arts in Boston, Massachusetts, owns a rocking chair that Maloof made from walnut wood.
- The armrests and the seat of the chair are sleek and contoured, and the back consists of seven spindle-like slats.

The student wants to describe the rocking chair to an audience unfamiliar with Sam Maloof. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. With its sleek, contoured armrests and seat, the walnut rocking chair in Boston's Museum of Fine Arts is just one piece of furniture created by American woodworker Sam Maloof.

B. Sam Maloof was born in 1916 and died in 2009, and during his life, he made a chair that you can see if you visit the Museum of Fine Arts in Boston.

C. Furniture designer Sam Maloof was a recipient of one of the John D. and Catherine T. MacArthur Foundation's "genius grants."

D. The rocking chair is made from walnut, and it has been shaped such that its armrests and seat are sleek and contoured.

Question 23

While researching a topic, a student has taken the following notes:

- In the early 1960s, the US had a strict national-origins quota system for immigrants.
- The number of new immigrants allowed from a country each year was based on how many people from that country lived in the US in 1890.
- This system favored immigrants from northern Europe.
- Almost 70% of slots were reserved for immigrants from Great Britain, Ireland, and Germany.
- The 1965 Hart-Celler Act abolished the national-origins quota system.

The student wants to present the significance of the Hart-Celler Act to an audience unfamiliar with the history of US immigration. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Almost 70% of slots were reserved for immigrants from Great Britain, Ireland, and Germany at the time the Hart-Celler Act was proposed.

B. Prior to the Hart-Celler Act, new immigration quotas were based on how many people from each country lived in the US in 1890.

C. The quota system in place in the early 1960s was abolished by the 1965 Hart-Celler Act.

D. The 1965 Hart-Celler Act abolished the national-origins quota system, which favored immigrants from northern Europe.

Question 24

While researching a topic, a student has taken the following notes:

- Soo Sunny Park is a Korean American artist who uses light as her primary medium of expression.
- She created her work *Unwoven Light* in 2013.
- *Unwoven Light* featured a chain-link fence fitted with iridescent plexiglass tiles.
- When light passed through the fence, colorful prisms formed.

The student wants to describe *Unwoven Light* to an audience unfamiliar with Soo Sunny Park. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Park's 2013 installation *Unwoven Light*, which included a chain-link fence and iridescent tiles made from plexiglass, featured light as its primary medium of expression.

B. Korean American light artist Soo Sunny Park created *Unwoven Light* in 2013.

C. The chain-link fence in Soo Sunny Park's *Unwoven Light* was fitted with tiles made from iridescent plexiglass.

D. In *Unwoven Light*, a 2013 work by Korean American artist Soo Sunny Park, light formed colorful prisms as it passed through a fence Park had fitted with iridescent tiles.

Question 25

While researching a topic, a student has taken the following notes:

- In 1999, astronomer Todd Henry studied the differences in surface temperature between the Sun and nearby stars.
- His team mapped all stars within 10 parsecs (approximately 200 trillion miles) of the Sun.
- The surface temperature of the Sun is around 9,800°F, which classifies it as a G star.
- 327 of the 357 stars in the study were classified as K or M stars, with surface temperatures under 8,900°F (cooler than the Sun).
- 11 of the 357 stars in the study were classified as A or F stars, with surface temperatures greater than 10,300°F (hotter than the Sun).

The student wants to emphasize how hot the Sun is relative to nearby stars. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. At around 9,800°F, which classifies it as a G star, the Sun is hotter than most but not all of the stars within 10 parsecs of it.

B. Astronomer Todd Henry determined that the Sun, at around 9,800°F, is a G star, and several other stars within a 10-parsec range are A or F stars.

C. Of the 357 stars within ten parsecs of the Sun, 327 are classified as K or M stars, with surface temperatures under 8,900°F.

D. While most of the stars within 10 parsecs of the Sun are classified as K, M, A, or F stars, the Sun is classified as a G star due to its surface temperature of 9,800°F.

Question 26

- In 2017, a research team led by Mary Caswell Stoddard determined the average lengths of eggs produced by various bird species.
- *Gygis alba* is a species of bird in the order Charadriiformes.
- *Gygis alba* eggs had an average length of 4.46 cm.
- *Gavia stellata* is a species of bird in the order Gaviiformes.
- *Gavia stellata* eggs had an average length of 7.22 cm.

Which choice most effectively uses information from the given sentences to emphasize a difference between the eggs of the two species?

A. A 2017 study compared the lengths of eggs produced by an array of different bird species, such as *Gygis alba* and *Gavia stellata*.

B. A 2017 study found that *Gygis alba* eggs had an average length of 4.46 cm, whereas *Gavia stellata* eggs were longer, with an average length of 7.22 cm.

C. The bird species *Gygis alba*, which belongs to the order Charadriiformes, and *Gavia stellata*, of the order Gaviiformes, were included in a 2017 study that compared the average lengths of their eggs.

D. Mary Caswell Stoddard led a research study that determined the average lengths of eggs, including those of *Gygis alba* birds (4.46 cm) and *Gavia stellata* birds (7.22 cm).

Question 27

While researching a topic, a student has taken the following notes:

- Chromosomes are cellular structures that contain genes.
- Genes carry critical instructions for determining an organism's physical traits.
- Members of the same species typically have the same number of chromosomes.
- The pineapple (*Ananas comosus*) and the melon (*Cucumis melo*) are species of fruits.
- The pineapple has fifty chromosomes.
- The melon has twenty-four chromosomes.

The student wants to specify how many chromosomes the pineapple has. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. The pineapple's chromosomes contain genes, which are critical to determining an organism's physical traits.

B. The pineapple (*Ananas comosus*) has fifty chromosomes.

C. The pineapple (*Ananas comosus*) and the melon (*Cucumis melo*) both have chromosomes, but the pineapple has more than the melon does.

D. The melon, a species of fruit, has twenty-four structures called chromosomes.

Question 28

While researching a topic, a student has taken the following notes:

- Tecozautla is a municipality in the state of Hidalgo, Mexico.
- Municipalities are governmental regions responsible for providing many public services to their residents.
- One service they provide is street lighting.
- Tecozautla covers an area of roughly 535 km².
- Hidalgo is divided into 84 municipalities.

The student wants to emphasize the size of Tecozautla. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. The municipality of Tecozautla in Hidalgo, Mexico, covers an area of roughly 535 km².

B. Providing street lighting is just one example of the public services that municipalities provide.

C. Tecozautla is one of 84 governmental regions, known as municipalities, across Hidalgo.

D. Tecozautla—a governmental region in the state of Hidalgo, Mexico—provides many public services to its residents.

Question 29

While researching a topic, a student has taken the following notes:

- Generally, an object will heat up when twisted.
- The twisting of an object is known as torsion.
- A 2019 study led by Zunfeng Liu and Ray Baughman tested the torsional heating of various fibers.
- When a 3-millimeter-thick sample of thermoplastic polyurethane (TPU) fiber was twisted, its average surface temperature increased by 6°C.
- When a 4-millimeter-thick sample of styrene-ethylene-butylene-styrene (SEBS) rubber fiber was twisted, its average surface temperature increased by 3.5°C.

The student wants to contrast the two samples. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. When the fibers were twisted as part of the 2019 study, the surface temperature of both samples increased.

B. In 2019, researchers studied the effect of torsional heating on various fibers, including samples of SEBS rubber and TPU.

C. Twisting an object will generally cause its temperature to increase, a process known as torsional heating.

D. The SEBS rubber sample used in the 2019 study was thicker than the TPU sample.

Question 30

While researching a topic, a student has taken the following notes:

- Pineapple is a fruit that contains ascorbic acid, an essential nutrient for humans.
- Every 100 grams (g) of pineapple contains 48 milligrams (mg) of ascorbic acid.
- Many animals can make ascorbic acid in their bodies, but humans cannot.
- Humans must get ascorbic acid from foods, including fruits and vegetables.
- Ascorbic acid is also known as vitamin C.

The student wants to provide an example of a fruit that contains vitamin C. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Humans cannot make ascorbic acid in their bodies, but they can get it from foods, such as fruits, for example.

B. Vitamin C, also known as ascorbic acid, can be found in pineapple as well as other fruits.

C. Since humans cannot make vitamin C in their bodies, they must get it from food.

D. Many animals can make ascorbic acid, which is also known as vitamin C, in their bodies, but humans cannot.

Question 31

While researching a topic, a student has taken the following notes:

- Texture analysis and historical analysis are two approaches to art criticism.
- Texture analysis examines how surfaces are visually represented in an artwork.
- Such an analysis of Giorgione's *Youth Holding an Arrow* might consider how the painting's blended colors make the subject's skin appear smooth in texture.
- Historical analysis considers the historical context in which a work was created.
- Such an analysis of Diego Velázquez's *Las Meninas* might consider how the painting's depiction of the artist with King Philip IV symbolizes art's historical ties to power.

The student wants to present historical analysis to an audience unfamiliar with the concept. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. A texture analysis of *Youth Holding an Arrow* might consider how the painting's blended colors make the subject's skin appear smooth in texture.

B. Texture analysis differs from historical analysis in that texture analysis examines how surfaces are visually represented in an artwork.

C. An approach to art criticism, historical analysis considers the historical context in which a work was created.

D. *Las Meninas*'s depiction of the artist with King Philip IV symbolizes art's historical ties to power.

Question 32

While researching a topic, a student has taken the following notes:

- Lighthouses send out crucial light signals to help ships and other watercraft navigate at night.
- Before automation, lighthouses were run by lighthouse keepers.
- Maria Younghans was the lighthouse keeper at Biloxi Light in Mississippi.
- She held this position from 1867 to 1918.
- Flora McNeil was the lighthouse keeper at Bridgeport Breakwater Light in Connecticut.
- She held this position from 1904 to 1920.

The student wants to emphasize the order in which the two lighthouse keepers began their careers. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. From 1867 to 1918, the nighttime waters of Mississippi were more navigable thanks to lighthouse keepers Flora McNeil and Maria Younghans.

B. Before automation, lighthouse keepers like Maria Younghans and Flora McNeil were crucial to ensuring safe navigation for watercraft.

C. Flora McNeil began her career as a lighthouse keeper years after Maria Younghans did.

D. Maria Younghans's career as a lighthouse keeper ended in 1918, whereas Flora McNeil's ended in 1920.

Question 33

While researching a topic, a student has taken the following notes:

- Hina Hanta is an online archive curated by the Choctaw Nation of Oklahoma.
- Hina Hanta means "bright path" in Choctaw.
- It features images of cultural artifacts relevant to the history of the Choctaw people.
- It includes a fanner basket (*ufko tapushik* in Choctaw) made from cane.
- It includes a robe (*nita anchi*) made from bear fur.

The student wants to specify the fanner basket's name in Choctaw. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. The Hina Hanta archive features cultural artifacts, such as a fanner basket and a robe, that are relevant to the history of the Choctaw people.

B. The cane fanner basket, which is included in the Hina Hanta online archive, is called an *ufko tapushik* in Choctaw.

C. Hina Hanta, which means "bright path" in Choctaw, includes a fanner basket in its archive.

D. The name of the online archive Hina Hanta means "bright path" in Choctaw.

Question 34

While researching a topic, a student has taken the following notes:

- Grimanesa Amoros is a Peruvian American artist well known for her LED light sculptures.
- Her sculpture *Uros Island* is made of smooth multicolored LED domes.
- It occupies 335 cubic feet of space.
- Her sculpture *Fortuna* is made of entangled blue and white LED tubes.
- It occupies 19,950 cubic feet of space.

The student wants to emphasize a similarity between *Uros Island* and *Fortuna*. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. The smooth LED domes of Grimanesa Amoros's *Uros Island* stand in contrast to the tangled LED tubes of *Fortuna*.

B. At 19,950 cubic feet in size, Grimanesa Amoros's *Fortuna* cuts a larger figure than the 335-cubic-foot *Uros Island*.

C. Grimanesa Amoros is the artist behind *Uros Island*—a sculpture made of smooth multicolored LED domes.

D. *Uros Island* is an LED light sculpture made by Grimanesa Amoros, as is *Fortuna*.